UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 07811

LH GENERAL PHYSICS

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. A rubber cord of cross sectional area A and initial length L is fixed at its upper end, and hangs vertically. Throughout this question, the fixed position of its upper end is adjusted so that the equilibrium position of its lower end is close to the Earth's surface. The cord has negligible mass and is made from a material having Young's modulus E.

   (a) Show that the extension (increase in length) e of the cord when a downward force F is applied to its lower end is given by LF/AE. **[4]**

   (b) The force in part (a) is provided by a mass m attached to the lower end of the cord. In equilibrium the extension is e. Find an expression for the net force on the mass when it is displaced a small vertical distance y from equilibrium. Assume that the gravitational field of the Earth has the constant value g. **[5]**

   (c) Show that the vertical position of the mass can undergo simple harmonic motion, and derive an expression for its period. Assume that the gravitational field of the Earth has the constant value g. **[8]**

   (d) A more precise calculation must allow for the fact that g depends on distance r from the centre of the Earth. Obtain an expression for g in terms of r, and hence $\frac{\mathrm{d}g}{\mathrm{d}r}$ at the Earth's surface. You may assume the Earth is a symmetric sphere having radius R_E, mass M_E, and gravitational field g_0 at its surface. **[7]**

   (e) Obtain an expression showing what happens to the frequency of oscillation when the effect in part (d) is taken into account. Consider the limits where $e \ll R_E$ and $e \sim R_E$. Comment on the feasibility of making an oscillator in the latter regime. **[6]**

2. Planets X and Y are a distance $d = 10^{12}\,\mathrm{m}$ apart in their common inertial frame. Observed in that frame, one event occurs at time $t_1 = 0\,\mathrm{s}$ on X, and a second event occurs at $t_2 = 3000\,\mathrm{s}$ on Y.

   (a) Calculate the space-time interval between these events. **[3]**

   Explain whether the event on X can be the cause of the event on Y. **[2]**

   What is the distance between X and Y, as observed from a spacecraft travelling from Y to X at $\sqrt{3/4}$ the speed of light? **[3]**

   Explain whether observers on this spacecraft conclude that there can be a causal relationship between the events. **[2]**

   (b) The event on planet X is associated with the instantaneous and isotropic emission of $N_\gamma = 10^{31}$ photons, each of energy $E_\gamma = 1.5\,\mathrm{eV}$. Calculate the number of these photons that arrive per square metre of planet Y, stating any assumptions that you make. **[5]**

   (c) A creature living on planet Y has a single light-sensitive organ which absorbs 10% of the photons that arrive at its $5\,\mathrm{mm}$ diameter circular aperture. The creature's response time is such that it is conscious of the number of photons absorbed in the most recent $0.1\,\mathrm{s}$ time interval. Background radiation on the planet has a Poisson distribution, and causes the organ to absorb photons at an average rate of 100 per second. Use statistical arguments to show whether the creature is likely to be aware of the event on planet X. **[10]**

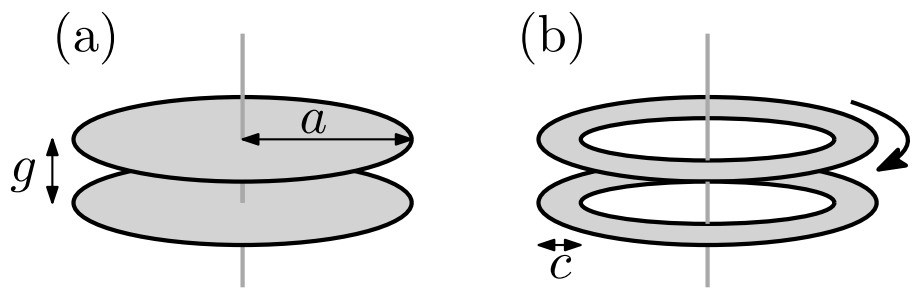
   (d) If the creature is to evolve to be almost certain to detect such events, compare quantitatively the effectiveness of increasing the size of its light-sensitive organ with increasing its 10% absorption. **[5]**

3. (a) Two circular metal discs, each of radius a are mounted parallel, with their centres spaced by a gap $g \ll a$ along an axis perpendicular to the discs and passing through their centres, see figure (a) below. A battery maintains a constant potential difference V between the discs.
Write down expressions for the capacitance between the discs, and the charge stored on each disc. **[5]**

(b) Both discs now have a circular hole cut from their centres, leaving two flat rings of width c, such that $g \ll c \ll a$, see figure (b) below. The top ring is made to rotate about the axis at a rate of f_1 revolutions per second. The potential difference between the rings is maintained at V.
Show that the capacitance between the rings is approximately $2\pi\epsilon_0 ac/g$, stating any approximations you make.
Write down an expression for the current constituted by the moving charge, assuming that the charges on the top ring rotate with the ring. **[8]**

(c) Obtain an expression for the magnetic field produced at the centre of the ring by the current calculated in part (b). Estimate, stating any simplifying approximations that you make, the magnetic flux that this current causes to pass through the lower ring. **[8]**

(d) The battery is now exchanged for an alternating voltage generator that applies a potential difference $V\sin(2\pi f_2 t)$ between the rings, where t is time. Obtain an expression for the EMF generated in the lower ring by the current in the upper ring.
Write down an expression for the resistance of the lower ring if it has resistivity ρ and thickness h.
Describe quantitatively the effect on the net magnetic field at the centre of the rings when current is allowed to flow around the lower ring. **[9]**



Formulae useful for General Physics. Vector notation is shown only in essential cases.

$$F = ma = \frac{\mathrm{d}p}{\mathrm{d}t}$$
$$v = u + at$$
$$v^2 = u^2 + 2as$$
$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = \boldsymbol{r} \times (m\boldsymbol{v})$$
$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F}$$
$$\tau = I\frac{\mathrm{d}\omega}{\mathrm{d}t}$$
$$I = mr^2$$
$$I_P = I_{cm} + Mr^2$$
$$F_c = mv^2/r$$
$$a_r = \frac{v^2}{r} = \omega^2 r$$
$$\boldsymbol{F'} = \boldsymbol{F} - 2m\boldsymbol{\omega} \times \boldsymbol{v} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$$
$$\boldsymbol{F} = -\boldsymbol{\nabla} U \quad \text{or} \quad -\frac{\mathrm{d}U}{\mathrm{d}x}$$
$$K = \frac{1}{2}I\omega^2$$
$$\omega_0 = 2\pi f_0 = \sqrt{\frac{k_s}{m}}$$
$$F_G = \frac{Gm_1m_2}{r^2}$$
$$U = mgh$$
$$\boldsymbol{E} = \frac{Q}{4\pi\epsilon_0 r^2} = -\boldsymbol{\nabla} V$$
$$V = -\frac{Q}{4\pi\epsilon_0 r}$$
$$\mathrm{d}\boldsymbol{B} = \frac{\mu_0 I\, \mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}}}{4\pi r^2}$$
$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0\left(\boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}\right)$$
$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$
$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}$$
$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$$
$$\Phi = \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A}$$
$$\epsilon = -\frac{\partial \Phi}{\partial t}$$

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q}{\epsilon_0}$$
$$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 \boldsymbol{I}$$
$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$
$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$$
$$\boldsymbol{D} = \epsilon_0 \boldsymbol{E} + \boldsymbol{P}$$
$$u = \frac{1}{2}\epsilon_0 E^2$$
$$u = \frac{1}{2}\frac{B^2}{\mu_0}$$
$$R = \rho L/A$$
$$P = VI = I^2R = V^2/R$$
$$C = Q/V; \;\; C = \epsilon_0 a/d$$
$$\Phi = LI$$
$$\Phi_2 = M_{21}I_1$$
$$U = \frac{Q^2}{2C}$$
$$U = \frac{1}{2}LI^2$$
$$PV = nRT$$
$$C = \frac{\mathrm{d}Q}{\mathrm{d}T}$$
$$\gamma = \frac{C_P}{C_V}$$
$$PV^\gamma = K$$
$$v = \sqrt{\frac{\gamma RT}{m}}$$
$$\mathrm{d}U = đW + đQ$$
$$S = k_B \ln \Omega$$
$$\mathrm{d}S \geq \frac{đQ}{T}$$
$$P(E_n) = \frac{g_n}{Z}\exp\left(-\frac{E_n}{k_BT}\right)$$
$$\dot{Q} = kA\boldsymbol{\nabla} T \quad \text{or} \quad kA\frac{\mathrm{d}T}{\mathrm{d}x}$$
$$\dot{Q} = \epsilon\sigma AT^4$$
$$\lambda_{\max}T = K$$
$$v_p = f\lambda = \frac{\omega}{k}$$
$$v_g = \frac{\partial \omega}{\partial k}$$

$$k = 2\pi/\lambda \quad ; \quad \omega = 2\pi f$$
$$Y = Y_0 \sin(kx - \omega t)$$
$$Y = Y_0 \exp i(kx - \omega t)$$
$$v = \sqrt{Y/\rho} \quad \text{or} \quad \sqrt{T/\mu}$$
$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \frac{v \pm v_{\text{rec}}}{v \pm v_{\text{src}}}$$
$$2d\sin\theta = n\lambda$$
$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \quad ; \quad \beta = v/c$$
$$x' = \gamma(x - ut) \quad ; \quad t' = \gamma(t - ux/c^2)$$
$$v' = \frac{v-u}{1 - uv/c^2}$$
$$p = \gamma m_0 v$$
$$E^2 = p^2c^2 + m_0^2c^4$$
$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \sqrt{\frac{1-\beta}{1+\beta}}$$
$$E = hf$$
$$p = \frac{h}{\lambda}$$
$$N = N_0\exp(-t/\tau) = N_0\exp(-\lambda t)$$
$$\Delta p \cdot \Delta x \geq \hbar$$
$$\Delta E \cdot \Delta t \geq \hbar$$
$$\hat{\boldsymbol{p}} = -i\hbar\boldsymbol{\nabla}$$
$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$
$$c = 1/\sqrt{\mu_0\epsilon_0}$$
$$n_1\sin\theta_1 = n_2\sin\theta_2$$
$$n = c/v$$
$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$
$$\sin\theta = 1.22\,\lambda/d$$
$$d\sin\theta = m\lambda$$
$$l = l_0(1 + \alpha\,\Delta T)$$
$$E = \frac{F/a}{\Delta l/l}$$
$$\gamma = F/d = W/a$$
$$p = \rho g h$$
$$F_{\text{up}} = (\rho_2 - \rho_1)gv$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.